

Computer Architecture Notes

What is Computer Architecture?

Computer Architecture is the design and organization of a computer system. It defines how the CPU, memory, and input/output devices communicate and execute instructions.

Main Components

1. CPU (Central Processing Unit)

The CPU is the brain of the computer.

Functions:

- Executes instructions
- Performs arithmetic operations
- Controls data flow
- Communicates with memory

Components:

- ALU (Arithmetic Logic Unit)
 - Control Unit
 - Registers
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2. Registers

Registers are very small and very fast memory locations inside the CPU.

Examples:

- Program Counter (PC)
- Instruction Register (IR)
- General Purpose Registers

Advantages:

- Fastest storage
 - Used during instruction execution
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3. Cache Memory

Cache is a high-speed memory located between CPU and RAM.

Levels:

- L1 Cache
- L2 Cache
- L3 Cache

Purpose:

- Reduce memory access time
 - Improve CPU performance
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4. RAM

Random Access Memory temporarily stores programs and data currently in use.

Characteristics:

- Volatile memory
 - Faster than storage devices
 - Used during execution
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Memory Hierarchy

From fastest to slowest:

Registers

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L1 Cache

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L2 Cache

↓

L3 Cache

↓

RAM

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SSD / HDD

CPU Pipeline

Pipeline stages:

1. Fetch

2. Decode
3. Execute
4. Memory Access
5. Write Back

Benefits:

- Improves throughput
 - Executes multiple instructions simultaneously
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Instruction Cycle

1. Fetch instruction
 2. Decode instruction
 3. Execute operation
 4. Store result
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Key Learning

- Understood CPU architecture
- Learned memory hierarchy
- Learned pipeline execution
- Studied instruction cycle